

User Manual: Garty's Architect (Arquitecto IA)

Your local control center for Artificial Intelligence

Welcome to Garty's Architect. This application has been designed to unify, simplify, and enhance the use of local Artificial Intelligence models. Forget about complex disconnected interfaces and code terminals; here is the ultimate bridge between your creativity, Ollama's structured reasoning, and ComfyUI's graphics engine.

Important Note: Garty's Architect is a professional orchestration tool. It assumes prior knowledge in managing and structuring generative AI.

1. Prerequisites and Recommended Hardware

To let the application unleash all its magic, you need to have Ollama and ComfyUI running on your computer (or local network). Being an unrestricted tool that allows loading Next-Gen models, the following hardware configuration is recommended for optimal performance:

- **Processor (CPU):** Modern multi-core processor (e.g., Intel i7-14700KF).
- **RAM:** 32GB of system memory to handle the handoff between vision, text, and image models.
- **Graphics Card (GPU):** RTX series card with 16GB of VRAM (e.g., RTX 5060) to run heavy architectures like Flux, LTX-Video, or Wan without bottlenecks.

2. Environment Preparation (Crucial)

Before starting Garty's Architect, you must ensure your base engines are correctly configured:

A. The Logical Core (Ollama) Ollama handles text processing, chat, and vision analysis (Default port 11434). The system strictly requires these three models to deploy its full potential:

- **SYS_LLM (llama3.1:8b):** The trunk logical engine. Without it, the application won't be able to generate or process prompts.
- **SYS_VISION (granite3.2):** The shadow analyst. Extracts vital information and tags from uploaded images to feed utilities like ReActor or IP-Adapter without barely consuming video memory.
- **VISION (qwen2.5-vl or Qwen 3.5 Vision):** The main descriptive analyst, capable of writing complex descriptions from photographs.

B. The Graphics Engine (ComfyUI) ComfyUI is the graphics engine for generating images and video (Default port 8188). The "Graphic Studio" advanced utilities require the following Custom Nodes to be installed (installing them via ComfyUI Manager is recommended):

- ComfyUI-ReActor (For Face Swap).
- ComfyUI_IPAdapter_plus (For style transfer).
- ComfyUI-Rembg (For background removal).
- ComfyUI-Advanced-ControlNet (To control image structure).

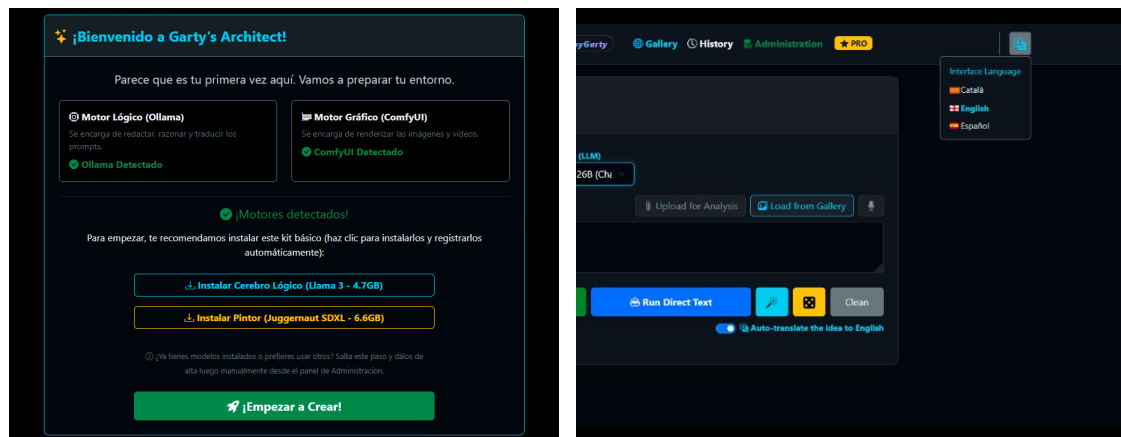
- ComfyUI-Impact-Pack (For ADetailer / Face Repair).
- ComfyUI_UltimateSDUpscale (For tile upscaling).

C. config.php File Configuration You must open the config.php file and set the absolute path to your ComfyUI models directory. This is useful if you have your heavy .safetensors files centralized on a secondary shared drive (for example, G:\ComfyUI\models).

3. First Steps: First Run and Multi-language

When launching Garty's Architect for the first time, the system will detect that your database is blank and will launch the Setup Wizard (Onboarding). During this phase, the application will verify the connection with the engines and guide you to register your first models.

Furthermore, the application is built from the ground up to adapt to you. From the settings, you can switch between English, Spanish, and Catalan. This change not only translates menus and buttons, but the internal engine will automatically search for AI instructions in the selected language so that it always communicates with you in your preferred language. From the administration panel, you can edit or create your own language.



4. Model and Personality Management

For the artificial intelligence to work for you, you must first organize its "brains". Garty's Architect allows you to manage all your models from a single centralized panel in the database.

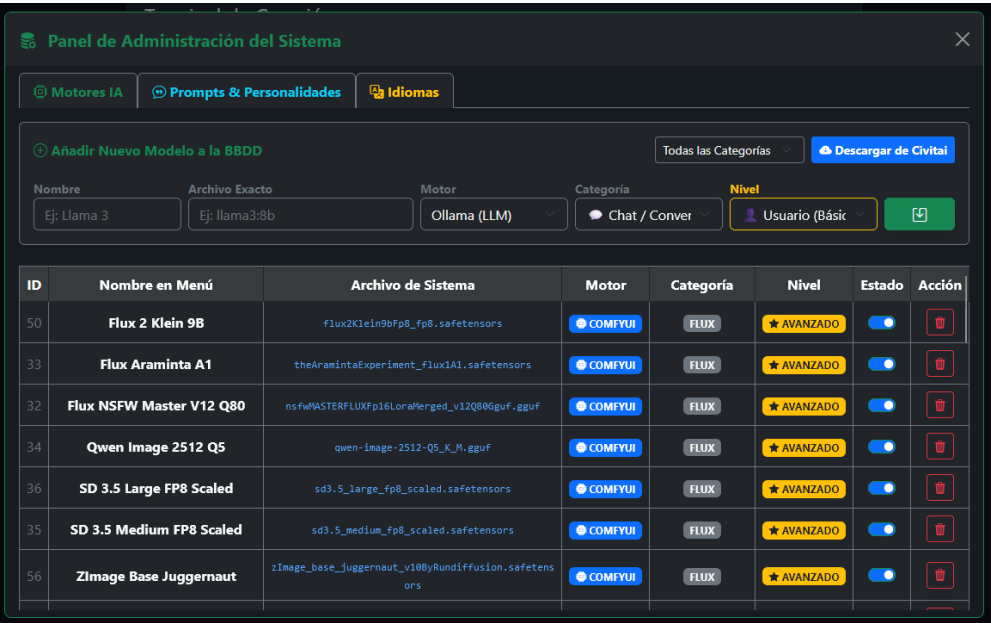
Model Types (The Muscle) When registering a model, you must provide its exact file name and assign it a category so the system knows when to use it:

- **[LLM] / [CHAT]:** Pure language models (e.g., Llama 3, Qwen, DeepSeek) to write, code, or converse.
- **[VISION]:** The main visual analyst (e.g., LLaVA). Capable of "seeing" images and describing them with a high level of literary detail.
- **System Models (SYS_VISION and SYS_LLM):** Lightweight models for internal use. For example, a SYS_VISION will work in the shadows to extract quick tags that feed tools like ControlNet or IP-Adapter, saving VRAM on your graphics card for the final generation.
- **Checkpoints:** The visual creation engines (SD 1.5, SDXL, SD 3.5, Flux). They dictate quality, text comprehension, and style.

- **LoRAs:** Small "patches" (styles, specific characters, clothing) that are added to the main Checkpoint.

💡 **Tip for LoRA Organization:** To keep your interface clean and avoid graphical errors caused by mixing incompatible architectures, organize your LoRA files by saving them into subfolders based on their corresponding engine (for example, create folders named Flux, ZImage, Qwen, or SD35). Garty's Architect is smart: it will read this folder structure and automatically filter the dropdown menu in the web interface, showing you *only* the LoRAs that are perfectly compatible with your currently selected image model.

- **Video:** Specific animation engines (Wan 2.1, LTX).



Prompts and Personalities (The Magic Dice) Next to the main text box, you will see a Dice icon. Clicking it will rotate the system through the different personalities or roles you have created (e.g., Programmer, Analyst, Creative Assistant).

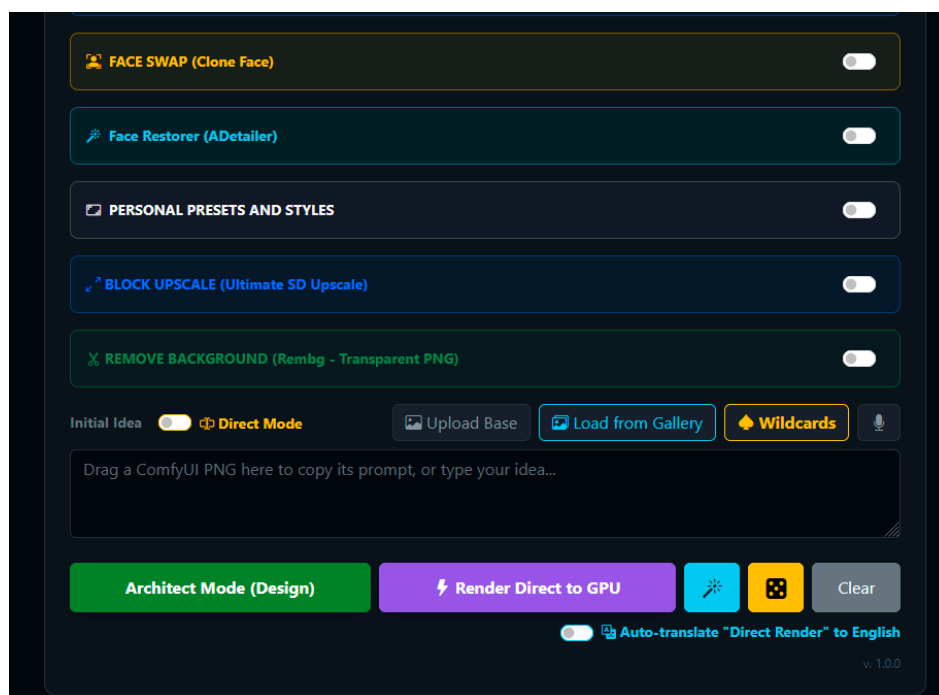
- **How it works:** The system automatically injects a hidden "System Prompt". If you choose "Programmer", the AI will know it should reply only with code, without introductions.
- **Advanced Configuration:** Each personality consists of a Type (e.g., vision_analyst), a Language, the Prompt Text (the secret instructions), and Parameters in JSON format. Through this JSON you can set, for example, that the vision analyst has a strict temperature of 0.0 so it doesn't hallucinate details, while a creative assistant has 0.8.

5. The Creation Interface: Flow and Assistance

The Magic Wand (Amplifier) If you have an idea but don't know how to formulate a good prompt, write something basic (e.g., "a cyberpunk city") and click the Magic Wand icon. Garty's Architect will send your idea to the system LLM and return it converted into a professional prompt, rich in detail, lighting, and composition, ready for the graphics engine.

Send to Generation vs. Direct to GPU Once you have your idea in the text box, you have two paths:

- **"Architect" Button (Send to Generation):** Moves your text to the graphic prompt boxes so you can review it. There you can add wildcards, adjust weights (e.g., (neon lights:1.4)), or configure extensions before rendering.
- **"Direct to GPU" Button:** Bypasses the review and sends the order immediately to the graphics card. It is ideal for fast and dynamic workflows.
- **The "Direct Mode" Switch (Manual Prompts)** If you've already done your homework and brought an exact prompt copied from external platforms (like Civitai), you don't need to go through the LLM's assistance. By activating the **Direct Mode** switch (located next to the Initial Idea label), the workspace will transform: the idea box and text tools will be hidden, giving way to two clean boxes for your **Positive Prompt** and **Negative Prompt**. Paste your texts there and click Render; the graphics engine will obey your instructions exactly, without any intermediaries.



6. Graphics Engine Settings and Styles

To master image generation, the advanced settings panel gives you absolute control:

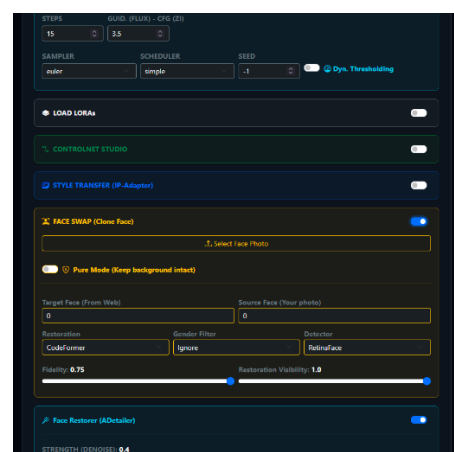
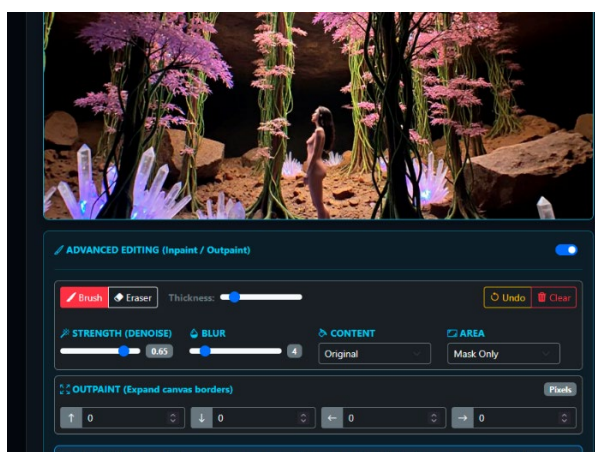
- **Steps:** How many iterations the AI does to clear the noise. Models like SDXL need 20-30 steps, while modern architectures like Flux or Turbo models require far fewer (4-8 steps).
- **CFG Scale:** Indicates how much the AI should obey your text versus its own freedom. (Usually 7.0 for SDXL, but lower, around 3.0, for Flux).
- **Guidance:** A specific parameter for Flux or SD 3.5 that separates prompt adherence from visual contrast, allowing hyper-realistic images without burning out the colors.

Styles and Presets Don't waste time rewriting your favorite settings. If you achieve the perfect combination (a structured prompt, a foolproof negative, and the exact settings), you can save it as a Preset. In future sessions, selecting that style from the dropdown menu will magically load all parameters.

7. The Advanced Graphic Studio (Extensions)

This is where Garty's Architect turns into a professional suite. By selecting a graphic model, you will be able to activate multiple "Switches" to shape the final result.

- **ControlNet (Structure):** Upload a sketch or a reference image. ControlNet extracts its structure (edges with Canny/Lineart, skeletons with OpenPose, or volumes with Depth) and forces the AI to generate a new image that respects that exact geometry.
- **IP-Adapter (Style Cloner):** Unlike ControlNet, IP-Adapter doesn't copy the shape, but the "vibe". It transfers the color palette, lighting, and atmosphere of your reference image to the new prompt.
- **ReActor (Face Swap):** Automated, high-precision face swapping. The AI will generate the image you requested and automatically graft your facial features onto the character, respecting the original scene's lighting and angle.
- **ADetailer (Face Restorer):** The magic button for perfect faces. Scans the finished image, detects if there are distant deformed faces or hands, crops them, redraws them at high resolution, and seamlessly integrates them back in.
- **Inpaint and Outpaint:**
 - **Inpaint:** Draw a mask over a specific area (e.g., a tie) and ask the AI to change only that, leaving the rest of the photograph intact.
 - **Outpaint:** Expands the margins of an image. The AI will invent what lies beyond the frame while respecting the landscape's coherence.
- **Rembg (Background Removal):** Cuts out the main subject from any image and removes the background with a single click, leaving it transparent (PNG).
- **Upscale (Tile Upscaling):** Enlarges images to gigantic resolutions without saturating VRAM by dividing the photo into small tiles, upscaling them, and stitching them back together.



8. LLM Module, Chat, and Vision Analyst

Garty's Architect is not just a graphic tool; it's a complete assistant.

- **LLM / Chat Mode:** Have fluid conversations, solve coding problems, or write complex texts. The system saves the history to keep the context alive.

Quick Generation Commands (Chat Shortcuts): You don't need to leave your conversation thread to bring your visual ideas to life. Right within the chat box, you can use special commands followed by your idea to summon the graphics engine in the background:

- **/img [your idea]** : The LLM you are chatting with will read your request, design a structured prompt optimized for the current graphics model, and send it silently to the GPU queue.
- **/gpu [your idea]** : Use this shortcut if you want to bypass the LLM's assistance and send your exact, literal text straight to the renderer.

In both cases, the system will process the request asynchronously. You can keep chatting and, when the image is ready, it will magically appear injected directly into the flow of the conversation.

- **Vision Analysis:** Select the [VISION] mode, upload a photo, and the application will call the vision_analyst personality to write a detailed and literary text describing every corner of the image.
- **Reverse Engineering (Drag & Drop):** Did you generate an incredible image and can't remember how? Drag any image generated by ComfyUI onto the application screen. Garty's Architect will read its hidden metadata and fill in all the boxes (prompt, seed, LoRAs, steps) so you can recreate it with a single click.

9. Performance and Asynchronous Workflow (The GPU Radar)

Generating complex images or videos takes time. Garty's Architect includes an asynchronous GPU Radar so you don't have to wait staring at a loading bar. An optimal environment will fully exploit these capabilities, although the system adapts to lesser setups thanks to lightweight system models.

- Upon clicking generate, the request is silently sent to the queue.
- You can continue browsing, chatting with the LLM, or reviewing your gallery.
- When the graphics card finishes its job, the system will notify you via a native Operating System Notification (if you are in another window) or with a green bubble (Toast) on the screen showing the thumbnail of your newly created artwork.

